RIVA
SYSTEM

RIVA — Real-time Intelligent Vehicle Assistant

Smart assistant that combines OBD diagnostics, live traffic cameras, and location services to optimize fuel, safety, and trip efficiency for drivers.

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Why RIVA Matters

Uncertain Fuel Needs

Drivers can't accurately predict fuel required for a trip across varying roads and traffic.

Hidden Vehicle Issues

Maintenance needs go unnoticed until breakdowns occur — costly and unsafe.

Route Inefficiency

Congestion increases travel time and fuel use without real-time guidance.

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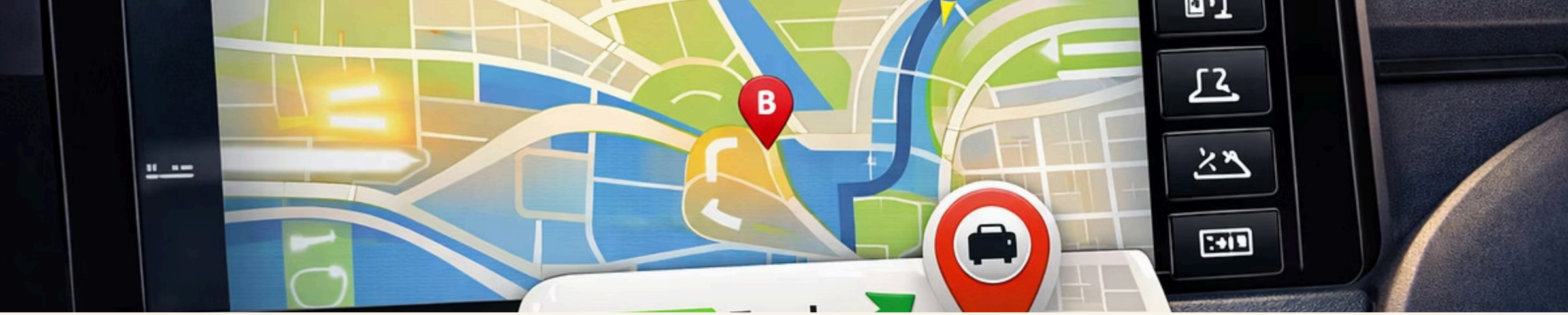

RIVA Solution Overview

RIVA unifies on-vehicle diagnostics, live traffic feeds, and location-aware services to deliver proactive fuel estimates, safety alerts, and route optimization — all in a mobile app.

Fuel Estimation

Trip-based fuel calculation using vehicle model, road type, and traffic.

OBD Diagnostics

Real-time engine and sensor monitoring, with predictive maintenance alerts.

Nearby Stations

Location-aware routing to nearest and best-priced fuel stations via Google Maps.

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Fuel Monitoring & Alerts

Continuously reads tank level and consumption from OBD. Sends staged notifications: low fuel warning, estimated remaining distance, and best refuel options based on location and price.

- Adaptive thresholds by driver preferences
- Estimated range calculation per route segment
- Push alerts with one-tap navigation

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Live Traffic Cameras & Route Optimization

Integrates live traffic camera streams and real-time congestion data to recommend faster, more fuel-efficient routes. Visual previews let drivers assess conditions before committing.

Benefit: Reduced idle time and lower fuel consumption from smarter routing decisions.

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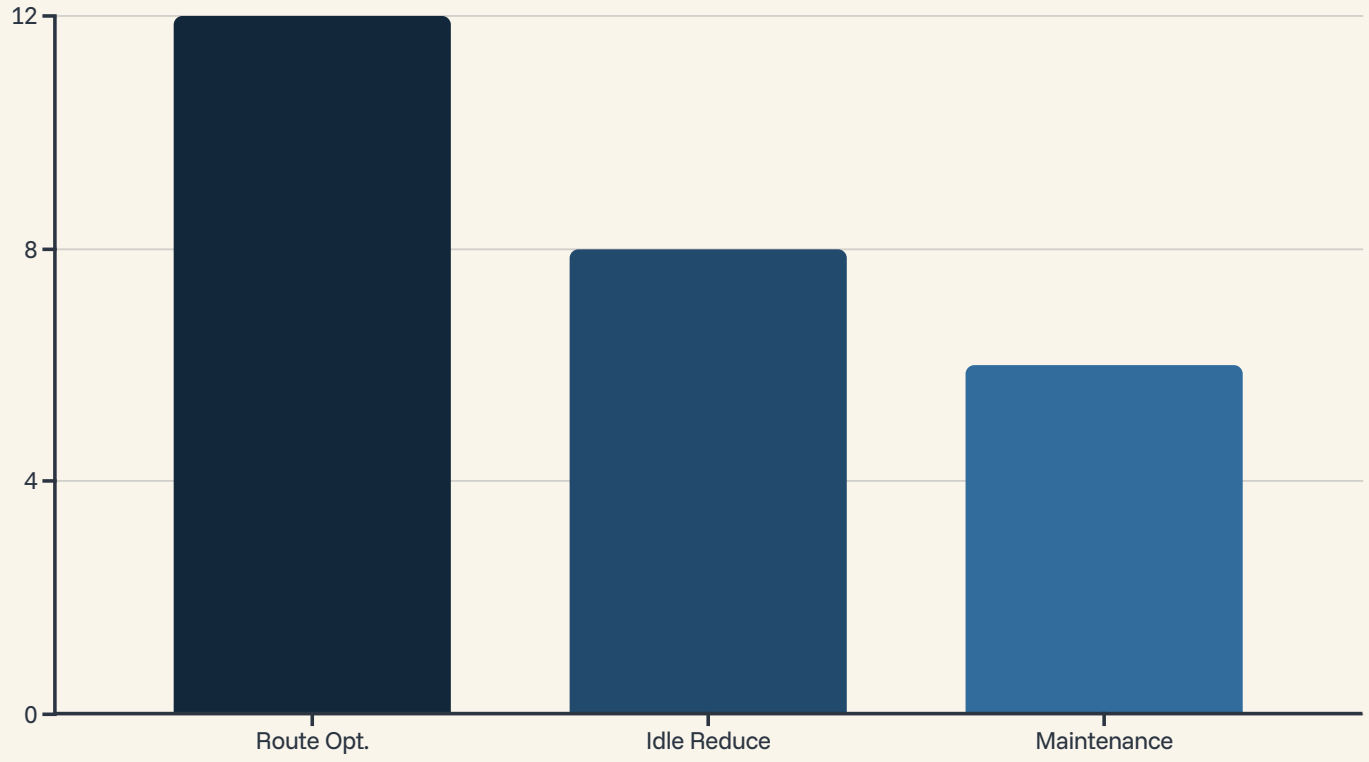

Projected Impact (Sample)

Early tests estimate double-digit fuel savings across typical urban commutes by combining smarter routing, reduced idling, and timely maintenance. Maintenance alerts reduce emergency repairs and downtime.

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OBD-II Integration & Diagnostics

1

Live Engine Data

Reads RPM, fuel trim, coolant temp, error codes — continuous health telemetry.

2

Predictive Alerts

Pattern detection flags potential issues before they escalate (e.g., misfire, sensor drift).

3

Service Guidance

Actionable advice: severity, nearest trusted service centers, and estimated repair impact on fuel efficiency.

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Monthly Reports & Driver Insights

Personalized monthly analytics summarizing distance, fuel used, cost, mpg, and top inefficiency drivers (idling, aggressive acceleration, route choices). Reports include actionable tips and estimated cost savings for improvements.

Reports help fleets and individual drivers make measurable, trackable efficiency improvements.

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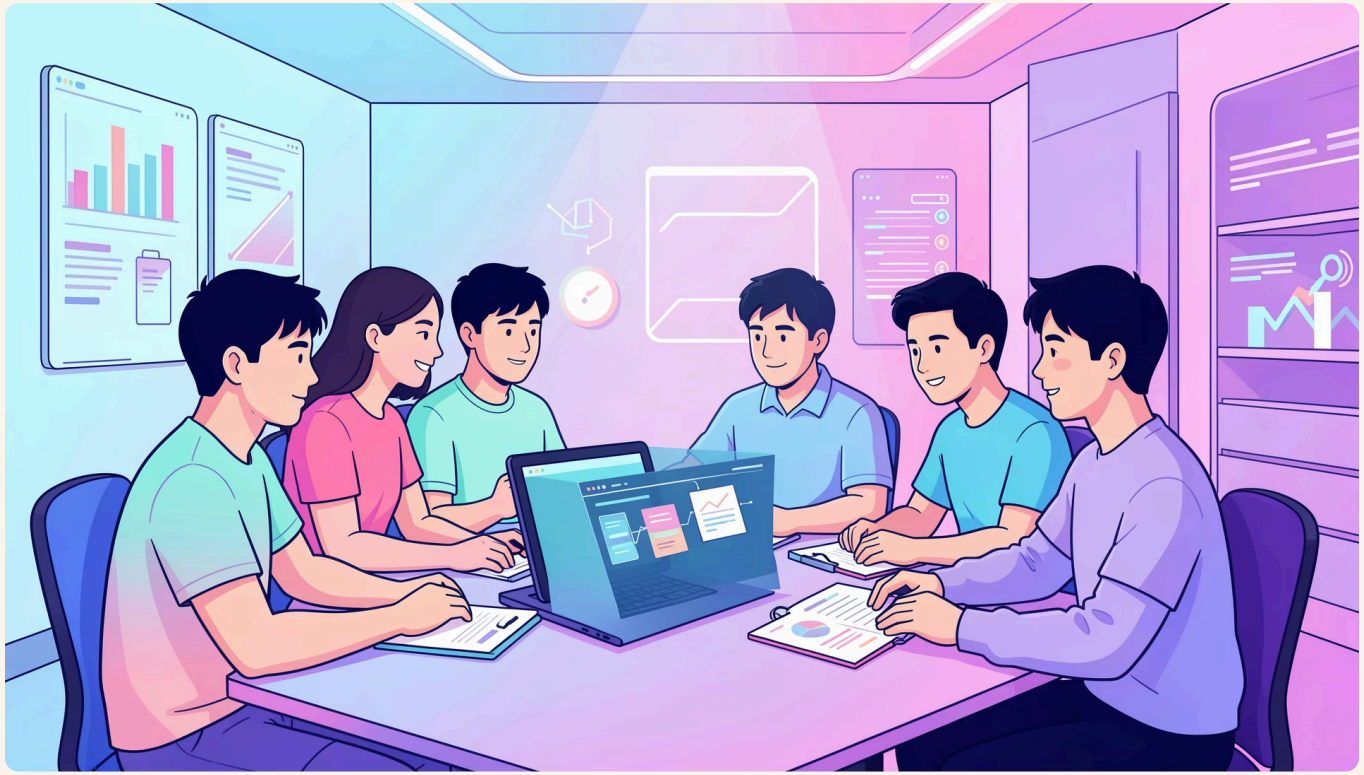

Technology Stack & Integrations

- Google Maps API — routing, POI, live traffic
- OBD-II interface — BLE/Wi-Fi dongles for real-time data
- Mobile apps — native Android & iOS with background processing
- Realtime processing & analytics — stream processing for live alerts
- Backend & storage — secure database for user and vehicle metrics
- Camera feeds — standardized ingest for traffic visualizations

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Next Steps & Implementation Roadmap

Pilot Integration

Connect OBD devices and Google Maps for a small fleet; validate fuel estimation accuracy.

Beta Release

Public beta for early adopters with live camera routing and monthly reports enabled.

Scale & Secure

Harden backend, expand integrations, and offer fleet-level dashboards and custom alerts.

Measure ROI

Track fuel savings, reduced downtime, and user satisfaction; iterate on features.

Request: approval to start a 3-month pilot with a 10-vehicle fleet. Expected outcome: validate 8–12% fuel savings and reduced unexpected maintenance.

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